MicroWeigh Automatic Tote Checking Scale

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Contents

1 Introduction

With the WeighTech MicroWeigh a combination of state-of-the-art technology with common down-to-earth basics creates a digital indicator that makes troubleshooting and actual maintenance repair so simple that anyone can be trained to make repairs on this indicator in just minutes.

1.1 MicroWeigh features

- High impact ABS alloy construction.
- Highly visible, easy-to-read display with adjustable contrast and backlight.
- Environmentally sealed touch-sensitive operator control panel.
- Standard units of measure include grams, kilograms, ounces, and pounds.
- RS-232 and Infrared communications are standard with RS-485 option available.
- Wireless data collection using a PDA with WeighTech ScaleTrax software.

1.2 MicroWeigh applications

- Standard weighing
- Tank or vat weighing
- Checkweighing (boxes, bags, and pieces)
- Bench and floor scales
- Batch weighing

2 Keypad Operation

The WeighTech MicroWeigh keypad is a watertight sealed touch sensitive sensor. The keys are actually sensitive to contact area, not force. Press lightly with the ball of your fingertip as though you were giving fingerprints. Best results come from using the ball of your finger, not the very tip. Most objects will not trigger the keypad–knives, screwdrivers, tools, etc. do not have enough surface area in contact with the key to register as a keypress. (You might get it to trigger with a medium sized conductive bolt head, if you have skin in contact with the bolt.)

One consequence of the design of the touch sensitive keypad is that it is sensitive to water streams. For this reason, WeighTech includes a unique "washdown mode" to prevent unwanted keypad activity during washdown/sanitation/cleanup intervals. When the indicator is in washdown mode, the indicator will weigh normally but the keypad is locked out.

To unlock the keypad, you must play follow the leader. One key will be lit. Press it. Another key will then light up. Press it. Continue until the indicator displays "Exit washdown". The indicator will require that you press five keys in a row correctly before it will unlock the keypad. Any wrong keypress will restart the counter back to five. The odds are extremely slight that random water splashing would ever be able to trigger the correct keys in the correct order to unlock the keypad.

3 Main Menu Items

3.1 "Power off"

Touch the enter key to select this menu item, which will power down the indicator. If the auto-on jumper is installed on the interface board, the indicator will immediately turn back on.

3.2 "Washdown"

This function puts the indicator in washdown mode to prevent inadvertent keypad activity. See the washdown section of this manual for more information.

3.3 "Totals"

This function leads to the totals submenu.

3.4 "Calibrate"

This function allows you to calibrate the scale. Refer to the calibration section of this manual for details.

3.5 "Setup menu"

Enter the setup submenu, where scale parameters can be viewed or set.

3.6 "Audit cfg"

Displays the audit counter for configuration. Every time a sealed scale parameter is modified this counter will increment by one. This setting is non-volatile (it will be retained even if the batteries go dead) and cannot be altered except by modifying an audited configuration parameter.

3.7 "Audit cal"

Displays the audit counter for calibration. Every time the scale is calibrated this counter will increment by one. This setting is non-volatile (it will be retained even if the batteries go dead) and cannot be altered except by performing a calibration.

3.8 “Tare”

Keypad entered tare: Touch the Enter key to set a new pushbutton tare by scrolling through digits one place at a time. Keypad tare values are entered in the current units, and are limited to be greater than gross zero weight and less than the indicator capacity. Entering a tare of zero will clear any existing tare from indicator.

4 How to Step Through Menus

From the main weight display, press the “Menu/Help” key. You are now in a menu, and the keys now have different functions:

Cancel	Help	Enter	Down	Up

- “Cancel” will back you out of the menu one level at a time.
- “Help” will display information about the current choice (option).
- “Enter” has various functions, depending on where you are in the menu.
- The “Down” key will scroll backward through the menu choices.
- The “Up” key will scroll forward through the menu choices.

4.1 Menus can contain several different items

An item with a “*” on the right end will do something when you press the enter key–something might be turn the indicator off, drill down into another menu, clear totals, or start a calibration routine. The item with a numeric value (scale capacity, for instance) at the right side of the display might allow you to change the number by pressing the enter key. An item with text (such as “on” or “off”) at the right side of the display might allow you to select from a list of options by pressing the enter key. Some items are just for reference and cannot be changed at all. Examples of reference items would be the firmware name and revision–these are set when the firmware is written and cannot be changed.

4.2 How to enter a number

Using the calibration routine as an example: Press the “Enter” key. The indicator display will show “Cal weight _” and the cursor will be blinking. The blinking cursor is the clue that you can enter an arbitrary number using the up, down, right, and enter keys. Pressing the up/down keys will scroll through the list (0 1 2 3 4 5 6 7 8 9 - .) in turn. When the desired number appears, press the right arrow (menu/help) key. The blinking cursor will advance one digit to the right, leaving your selected number in place. Continue this sequence until the desired numeric value is visible on the display. Press the “Enter” key to accept the value, or the “Cancel” key to abort.

Example: Enter a calibration weight of 25 pounds

- Start with the indicator at a normal weight display (“ 0.00 lb”)

- Press the “Menu/Help” key
- Scroll through the main menu using the up or down arrow keys until “Calibrate *” is displayed on the indicator
- Press the “Enter” key to start the calibration routine
- The indicator may display “Password” if a calibration password is required. If so, enter it (default calibration password is “Zero” “Zero” “Zero”)
- The indicator should now be displaying “Cal Weight” and a blinking cursor.
- Press the up arrow key. The display should now show “Cal weight 1”
- Press the up arrow key again. The display should now show “Cal Weight 2”
- Press the right arrow key to accept the first digit (2) and advance the blinking cursor to the next digit. The indicator should display “Cal weight 2_”
- Press the up arrow key five times to select a 5 as the second digit. The indicator should now display “Cal weight 25”
- Press the “Enter” key to accept 25 pounds as a calibration weight.
- The indicator will display “Cal–zero weight”. Press the “Cancel” key to abort the calibration process.

4.3 How to select from a list

This is very much like stepping through a menu. Some settings (such as displayed resolution) must be limited to one of several predetermined values. To edit one of these settings, press the “Enter” key. The currently selected value will move from the far right of the display to the left. This indicates that you may use the up and down arrow keys to scroll through a list of possible values for this setting. Once you’ve selected a value for the setting, press the “Enter” key to complete the selection process. As always, pressing the “Cancel” key will cancel the selection and restore the setting to the previous value.

5 General Scale Operations

5.1 Scale on procedure

Touch the “Zero / On” key. Indicator will come on and display will read “MicroWeigh by WeighTech” and then continue to the weigh mode. At this point the scale is ready for product or operator input.

5.2 Scale off procedure

To turn the scale off touch the “Menu / Help” key. The indicator will display “Power off *”. At this point touch the “Print / Enter” key and scale display will go blank, and the indicator will be off. (If the auto-on jumper is installed on the interface board, the indicator will immediately power up.)

5.3 Zero procedure

To zero the indicator touch the “Zero / On” key and the indicator will take a new zero. If the current weight reading is unstable, under capacity, or over capacity, no new pushbutton zero will be established.

5.4 Units procedure

To change the units of measure touch the “Units / Cancel” key. The units will change between pounds, kilograms, grams and ounces (assuming all the units are enabled in the “Parameter” menu) each time that the key is touched.

5.5 Tare operation

Press and hold the tare button to establish a pushbutton tare reference. If a valid tare is established, the indicator will switch to the net weight display. If the gross weight is equal to or less than gross zero, any existing tare value will be cleared, the display will show “Tare cleared” for about one second, and the display will revert to gross weight display.

Toggle between net and gross display modes by touching the “Tare” button. If no tare reference has been established, the indicator will not switch to net weight mode.

An arbitrary tare weight can be entered from the tare setting in the main menu (keypad tare). Scroll and select digits one at a time to enter the desired value. The indicator will not accept a keypad tare value in excess of scale capacity, or less than zero. Entering a value of zero will clear any existing tare and return the indicator to the gross weight display mode. Units for the entered weight is the same as the currently displayed units. (To enter a six pound tare, be sure that the display is showing weight in pounds before entering the keypad tare.)

6 Setup Procedure

Once the wiring and plumbing is complete, apply power to the indicator and turn off the drive motor. Set the scale capacity and calibrate the scale. Go to “Setup menu”/“Test menu” and step through each of the output tests (stop gate, weighbridge, and knockoff). Check that each test works the correct air cylinder. Scroll to “Sensor” and check that the display shows “No tote” when the optical sensor beam is clear and “Tote” when the beam is blocked. (Flip the “Setup menu” / “Tote settings” / “Opto mode” setting if it’s reversed.)

If the system is to knock off rejected boxes, set “Setup menu” / “Tote settings” / “Knockoff” to on and “Tolerance” to off. Boxes with a net weight less than “Tol low” or net weight greater than “Tol high” will be rejected and knocked off.

Another option would be to set “Knockoff” to off and “Tolerance” to on. In this case boxes with a net weight outside the tolerance band will stay on the weighbridge until the operator corrects the box weight. (Set “Avg Piece” to the average piece weight, in pounds, and turn on “Prompt” to show the operator how many pieces must be added or removed from the box.)

Go into “Setup menu” / “Tote settings” and set “Detect” to “Opto” to use the optical sensor as the box detection method. (Use “Weigh” if the machine does not have the optional opto box sensor.) Set “Box edge” to “Lead” if you’d like to time off the front edge of the box, or “Trail” if you’d like to time off the back edge of the box. Leading edge detection is preferred if the boxes are about the same size. Trailing edge detection might work better if the boxes vary in size.

Power on the drive motor so that the belt is running. Put the system in the cycle on mode. Feed one box at a time and tune the “Setup menu” / “Tote settings” / “Sense TMR” so that the stop gate comes up right behind the box. Next, feed boxes one at a time and adjust the “Delay TMR” so that the box is approximately centered on the weighbridge when it lifts the box. If you’re using the knockoff, tune the “KO Delay” and “KO Duration” to cleanly kick the rejects off the line. Add a little “Lift TMR” if the system appears to be getting a stable weight before the box is completely lifted. Finally, tune the “Eject TMR” so that the box stop comes down after the box has left the weighbridge area. (The system knows that the box has left the weighbridge when the weight drops below “Eject wgt”.)

If the system has an infeed box pincher in front of the box stop, tune the “Pinch TMR”. This is the delay between the box stop going down and the pincher releasing the second box. Set it so that there is a consistent gap between boxes, and that gap is wide enough for the box stop to reliably stop the second box.

If you’d like the system to automatically zero between boxes, turn on “Setup menu” / “Tote settings” / “Autozero”.

7 Calibration Procedure

7.1 Entering the calibration menu

With the indicator on and displaying weight, touch the “Menu / Help” key. The display will read “Power off *”. Use the up / down arrows until the display reads “Calibrate *”. Touch the “Print / Enter” key and the display should then show “Password”. At this point key in the calibration password. (The default calibration password is “Zero” “Zero” “Zero” .)

7.2 Keying in cal weight

The display will show “Cal weight _” and the cursor will be blinking. Using the up, down, and right keys to enter the size of your calibration weight in pounds (i.e. 1, 2, 5, or 10). Press “Enter” to accept the cal weight, or “Cancel” if you make a mistake.

7.3 Calibration example

(Entering a 25.00 lb cal weight value.) The blinking cursor is the clue that you can enter an arbitrary number using the up and down keys. Pressing the up/down keys will scroll through the list (0 1 2 3 4 5 6 7 8 9 - .) in turn. When the desired number appears (2), press the right arrow “Menu / Help” key. The blinking cursor will advance one digit to the right (2 _), leaving your selected number in place. Continue this sequence until the desired numeric value is visible on the display (25_) (25._) (25.0_) (25.00). Press the “Enter” key to accept the value, or the “Cancel” key to abort.

7.4 Establishing a zero

The indicator will display “Cal–zero weight”. Clear the weighing platform of any foreign objects and once all vibration has died out, press the “Enter” key. Make sure that the platform is not disturbed during this process. Indicator will display “Zeroing...” as it takes an average reading of the zero offset weight (about three seconds).

7.5 Accepting a cal weight

The indicator will then display “Cal–add weight”. Add weight to the weighing platform (the weight should be the same amount as the keyed in cal weight) then touch the “Enter” key. The indicator will display “Scaling...” for about three seconds as it performs internal calculations. Finally, the indicator will display “Cal done” for about one second once the calibration cycle is complete.

8 Scale Parameters

To get to the parameters touch the “Menu/Help” key (indicator will display “Power off *”). Use the up or down arrows until the indicator displays “Setup Menu”. Touch the “Print/Enter” key and the indicator will prompt for a password. The password for this step will be as follows: starting from the left side of the keypad touch each key in turn from left to right. After entering the password the indicator will display “Parameters *”. At this point touch the “Print/Enter” key to access the parameters. Use the up and down arrows to scroll through and view each parameter.

8.1 “Units”

This parameter controls the setup unit of the indicator. Select from pounds (lb), kilograms (kg), grams (g), and ounces (oz). Once set, the indicator capacity, resolution, and calibration weights will be entered in this unit. The units parameter is both sealed and audited.

8.2 “Capacity”

Capacity sets the maximum capacity of the indicator, in setup units. This parameter is both sealed and audited. Factory default is 0, which must be changed before the indicator will weigh.

8.3 “Resltn”

Parameter that sets the resolution of the indicator. Resolution is limited to values available on the scroll list. Resolution is set in terms of the setup units. This parameter is both sealed and audited.

8.4 “Stability”

This parameter controls how many consecutive weight readings are required to be within the motion sense band before the weight indication is considered to be stable. The indicator reads the analog input 7.5 Hz (7.5 times per second), so the default setting of four requires about a half second of stable weight. Either the net or gross light will come on when the weight is stable. This parameter is both sealed and audited.

8.5 “Motion sns”

Amount of motion, in divisions, allowed before the weight is considered unstable. Default is one division. This parameter is both sealed and audited.

8.6 “AZT”

Auto zero tracking on/off. This parameter is neither sealed nor audited. When on, stable weights within the “AZT band” of zero will automatically rezero the scale.

8.7 “AZT band”

Amount of weight, in divisions, that can be automatically zeroed out at one time. Default is 1 division. Parameter is sealed and audited.

8.8 “Calibrate”

This function starts the indicator calibration routine. It is sealed and audited. Refer to the calibration section of this manual for details.

8.9 “IZ set”

When this parameter is on, the indicator will attempt to establish a new initial zero every time the indicator powers on. HB44 limits the amount of weight that can be initially zeroed to 20% of scale capacity. (This initial zero does not reduce the indicator capacity.) This parameter is both sealed and audited.

8.10 "lb units"

Select on/off to enable or disable the pounds (lb) units when the Unit key is pressed in weighing mode. This parameter is both sealed and audited.

8.11 "kg units"

Select on/off to enable or disable the kilograms (kg) units when the Unit key is pressed in weighing mode. This parameter is both sealed and audited.

8.12 "g units"

Select on/off to enable or disable the grams (g) units when the Unit key is pressed in weighing mode. This parameter is both sealed and audited.

8.13 "oz units"

Select on/off to enable or disable the ounces (oz) units when the Unit key is pressed in weighing mode. This parameter is both sealed and audited.

8.14 "Defaults"

Restore all configuration parameters to factory default. This function is sealed and audited. Restoring factory defaults will require that the indicator be calibrated and reconfigured before it will weigh.

9 Display Messages

9.1 General warning/error messages

The following warning and error messages may appear at any time that the display is showing the current weight. They will not be visible when the indicator is displaying a menu item.

"Setup required" The indicator is still set to factory defaults, and will require configuration before entering service. This message will clear once the scale capacity (in "Setup Menu"/"Parameters") has been set.

"Cal required" The indicator has not yet been calibrated. This message will clear once the scale has been calibrated.

"Excite Shorted!" The measured load cell excitation voltage has been below 1V for more than one second. To prevent any damage to the indicator or load cell, the excitation supply has been disabled. Double check the load cell cable and connections to the interface board, especially at the terminals marked "EX+" and "EX-". The indicator must be turned off and back on again to clear this message.

“ADC Full Scale” The analog to digital converter reading the load cell went to full scale positive or negative for more than a second. This is usually caused by faulty wiring, or in severe cases, a seriously damaged load cell. Check connections, especially on the terminals marked “S1+” and “S1-”. This warning will clear automatically when the load cell readings come back into the normal range.

“IR locked.” Access to the setup menus is locked out. The “IR lock” setting in the parameter menu is turned on and it’s been more than one minute since the Palm was used to pull settings or totals. To unlock, use a Palm to pull totals or settings. You will then have 60 seconds to access the setup menus. If you do not wish to lock out the setup menus, turn off “IR lock”.

9.2 Calibration warnings

The following warning messages may appear when entering the calibration routine.

“Check load cell” This warning will only appear when entering the calibration routine. It indicates that the load cell reading is past full scale positive or negative. Calibration will not be allowed in this case. Refer to “ADC Full Scale” warning for troubleshooting information.

“Check excite” The load cell excitation supply was measured at less than 4.0 volts. Check the load cell and wiring, especially the terminals marked “EX+” and “EX-”. This warning can also be caused by excessive load cell current drain, such as from more than four load cells tied together, or a load cell with damaged strain gauges.

“Check ex- wire” Both load cell signal outputs were measured at more than 4.0 volts. This usually indicates that there’s a poor connection at the “EX-” terminal, or that the load cell excitation wiring has gone open (such as from physical damage to the cable).

“Check ex+ wire” Both load cell signal outputs were measured at less than 1.0 volts. This usually indicates that there’s a poor connection at the “EX+” terminal, or that the load cell excitation wiring has gone open (such as from physical damage to the cable).

“Check signals” The difference between the two load cell signal outputs was measured at more than 0.2 volts. This much imbalance between the two signals is usually caused by either incorrect wiring (such as swapping an excitation and signal wire), or by a severely bent load cell. Check connections, try swapping pairs, or replace the load cell.

“Check sig+ wire” The positive load cell signal was measured at either less than 1.5 volts or more than 4 volts. Check wiring at terminal marked “S1+”, check that the load cell signal and excite pairs are correct, and finally consider that the load cell may be damaged.

"Check sig- wire" The negative load cell signal was measured at either less than 1.5 volts or more than 4 volts. Check wiring at terminal marked "S1-", check that the load cell signal and excite pairs are correct, and finally consider that the load cell may be damaged.

10 Menus

10.1 Main level

Power off	Turn off the indicator
Production	Enter the production submenu
Washdown	Disable keypad to prevent false keypresses during washdown
Totals	Display the totals and statistics
Calibrate	Enter quick calibration routine
Setup menu	Bunch of stuff...see below
Audit cfg	Number of times an audited config parameter has been changed (HB44)
Audit cal	Number of times indicator has been calibrated (HB44)
Tare	Current tare weight

10.2 Setup menu

Parameters	Scale settings
Tote settings	Tote settings
I/O test	Module test menu
Comm menu	RS-485 communications menu
Info menu	Troubleshooting features
Clock	Set time/date
Contrast	Control display intensity

10.3 Production

Cycle on	Start automatic weighing cycle
Cycle off	Stop automatic weighing cycle (stop and tote lift down)
Clear stats	Clear current totals/stats
Tol low	Low side of acceptable weight tolerance band
Tol high	High side of acceptable weight tolerance band

10.4 Total menu

* *cnt*	Total number of items that have been weighed
* *tot*	Total weight of all items
* *avg*	Average weight of all items
* *min*	Minimum weight of all items
* *max*	Maximum weight of all items
* *dev*	Standard deviation of all items
OK cnt	Total number of acceptable items
OK tot	Total weight of acceptable items
OK avg	Average weight of acceptable items
OK min	Minimum weight of acceptable items
OK max	Maximum weight of acceptable items
OK dev	Standard deviation of acceptable items
Rj cnt	Total number of items that have been rejected
Rj tot	Total weight of rejected items
Rj avg	Average weight of rejected items
Rj min	Minimum weight of rejected items
Rj max	Maximum weight of rejected items
Rj dev	Standard deviation of rejected items
Tamper	Number of boxes that may not have been weighed
Clear stats	Clear current totals/stats–you will be asked to confirm by pressing "Enter" a second time

10.5 Parameters

Capacity	Scale capacity, in setup units
Resltn	Scale resolution, in setup units
Stability	Number of consecutive readings required for stability
Motion sns	Number of divisions allowed before weight is considered unstable
Disp rate	Control number of updates: larger numbers are slower
Filter fact	Weighing filter speed: range of 0-0.9. Larger numbers make the filter slower, but weights are more stable
AZT	On/Off: Autozero tracking on/off, only affects weights near zero
AZT band	Amount of weight (in divisions) that can be zero tracked out
Calibrate	Start calibration routine
IZ set	Set initial zero at power up (default to off)
lb units	On/Off: Enables the units toggle key to include pound units
kg units	On/Off: Enables the units toggle key to include kilogram units
g units	On/Off: Enables the units toggle key to include gram units
oz units	On/Off: Enables the units toggle key to include ounce units
Tare	Enable/Disable: controls pushbutton tare option
Address	Scale communications address/ID number
Cntst	Enable or disable the three key quick contrast adjustment
IR lock	When turned on, the Palm is required to access the setup menus
Defaults	Restore scale to factory default settings (all settings will be lost!)

10.6 Tote settings

Tolerance	Set to on to manually make weight
Tol low	Low side of acceptable weight tolerance band
Tol high	High side of acceptable weight tolerance band
Knockoff	Set to on to knock off rejected boxes
KO delay	Delay from box down to knockoff activate
KO duration	Duration of knockoff
Sense wgt	Weight required to activate the tote lift cycle (weigh detect only)
Eject wgt	Weight required to consider the tote ejected
Prompt	Set to on for a piece adjustment prompt while making weight
Avg piece	Average piece weight, in pounds (use with prompt setting)
Autozero	Turn on to automatically zero weight after ejection
Sense TMR	Delay between box sense and box stop up (in seconds)
Delay TMR	Delay between box stop up and tote up (in seconds)
Lift TMR	Delay between tote up and actually weighing the tote (in seconds)
Weigh TMR	Not used
Prompt TMR	Not used
Eject TMR	Time between box not sensed (after ejection) and box stop down (in seconds)
Pinch TMR	Delay from box stop down until infeed box pincher releases the next box
Box edge	Lead/Trail: select active edge of box (opto detect only)
Opto mode	Select light or dark operate opto sensor
Detect	Select whether to detect boxes from weight or an optional opto sensor
Force net	Turn on to force net weight and disable gross
RS232	Select RS-232 datastream (GK5 or Plgrm 1)
Verify drop	When on, press enter after making weight to drop the weighdeck

10.7 I/O test menu

Stop gate	Press to toggle box stop gate
Weigh bridge	Press to toggle tote lift (weighbridge)
Knockoff	Press to toggle knockoff section
Sensor	Tote/No Tote: live display of opto sensor

10.8 Comm menu

Rx	Number of bytes received over RS-485
Tx	Number of bytes transmitted over RS-485
In	Number of packets received over RS-485
Out	Number of packets transmitted over RS-485
CRC	Number of corrupt packets received over RS-485
Err	Number of bytes received with errors

10.9 Info menu

Sensor	State of opto sensor
232 in	Bytes received over RS-232
ADC	Raw counts display from analog to digital converter
Offset	Calibration zero offset, in raw counts
App	Name of firmware app (*tote_check_4*)
Build	Software revision info (*Build 37*)
Date	Date firmware was compiled (*01/23/2017*)
Time	Time of firmware compilation (*09:59:51*)
Batt	Current power supply/battery input voltage, in V
Batt disp	Not used
S1+	Load cell #1 positive signal voltage (should be about half of excite voltage with good load cell)
S1—	Load cell #1 negative signal voltage (should be almost exactly the same as S1+ voltage)
Excite	Load cell excitation voltage (should be about 4.5V)
Deadload	Display current estimated load cell deadload
TimeTest	
232 audit	
IZ autoset	
IZ	
Debug msg	On/Off: Turn this parameter on for more extensive messages during boot and dump cycle
Bootload	WeighTech use only

11 Detailed Menu Item Descriptions

11.1 Main menu level

Power off Turns off the MicroWeigh indicator. This will not turn off the remote keypad or the label printer.

Washdown Disable the keypad to prevent the touch sensitive keypad from registering false keypresses during a washdown period. To enable the keypad, play follow the leader and press the lit keys in order. You must get five in a row to exit washdown mode. Washdown mode only disables the five touch sensitive keys on the indicator; the remote keypad is not disabled in washdown mode.

Totals Enter the totals menu. From inside the menu, totals can be viewed and cleared. There are totals (weight and case count) for each shift and each product code. The totals are also available for download to a Palm through the front panel IR interface at any time. The indicator will ask for confirmation (“Sure?”) before clearing totals. Touch the enter key to confirm and clear, or touch the cancel key to abort without clearing totals.

Calibrate Quick calibration routine. The password required to enter the calibration routine is to touch the Zero/On key (far right side of keypad) three times. The indicator will first prompt for a calibration weight and the calibration weight units, then prompt for all weight to be removed, and finally for the calibration weight to be added. Before calibrating, several tests are run on the load cell and load cell connections. If any of these load cell tests fail, a warning message will displayed before the calibration weight prompt.

Setup menu Enter the setup submenu, where the configuration settings and test functions are located. This menu is password protected to prevent unqualified personnel from accessing the settings. The password is to touch each of the five keys in order, from left to right. ("Units", "Menu", "Print", "Tare", "Zero").

Audit cfg Displays the number of times an audited configuration parameter has been changed. There is no way to reset this counter in the field.

Audit cal Displays the number of times the indicator has been calibrated. There is no method to reset this counter in the field.

Tare Displays the current tare weight. A keypad entered tare weight can also be set, if desired.

11.2 Setup menu

Parameters Enter the parameter menu, where all the important (nonvolatile) settings are located. A word of warning—the "Default" option exists to return the settings to the factory defaults. It will prompt for confirmation, but once you've defaulted the settings any changes you've made are gone forever. If you don't know why you'd need to return to factory defaults, you don't.

Info menu Enter the info menu, which contains several interesting and useful nonadjustable parameters, such as the current firmware name and revision.

Clock Display current date and time. Also allows setting the real time clock. (The MicroWeigh controller real time clock is battery operated, so it will keep time even while the controller is powered off.) To set the clock, enter the current date at the "Date" prompt in mm.dd.yy format. (The exact format is flexible; for example, March 1 2006 could be entered as "4.1.6", "04-01-2006", or "4.01-06".) Enter the current time in 24 hour hh.mm.ss format at the "Time" prompt. (Again, it's flexible enough to accept "14.0", "14-00-00", or "14" for 2PM.) Once the current date and time have been entered, scroll to the "Set clock" option and touch the "Enter" key. The indicator will then set the clock and confirm by displaying "Clock set".

Contrast Allows adjustment of the display intensity. Use the up and down arrows to scroll until the display contrast is nice and crisp, then press the Units/Cancel key to exit. This setting is retained through power cycles. There's an emergency contrast adjustment that's handy if the display is really washed out and hard to read: touch and hold the Units/Cancel and Menu/Help keys, then touch either the up or down key to adjust contrast. If you can't tell whether to go up or down, touch the same key repeatedly until the display is easy to read. (It'll loop around, so it doesn't really matter if you use the up or down key.)

11.3 Parameters

Capacity Maximum capacity of scale, in setup units. Must be set during initial installation to remove the "Setup Required" warning message from the display.

Resltn Displayed weight resolution, in pounds. Default is 0.05 pounds.

Stability Number of weight readings required to be within "Motion sns" divisions before the weight is considered stable. Default settings is 4, which will require about 1/2s for the scale to stabilize. (The load cell is read 7.5 times per second.)

Motion sns Size of band, in divisions, that weights must fall into to be considered stable. See "Stability" .

Filter Filter control: Larger numbers will give a slower, more stable weight reading. Range is 0.01 to 0.99. Default value is 0.2.

AZT (Automatic Zero Tracking) On/Off: when on, the scale will automatically zero out minor weight fluctuations need zero weight.

AZT band Number of divisions that AZT can automatically zero out.

Calibrate Perform a load cell test and calibration. (Feel free to enter the routine to do a load cell test, and touch the cancel key to exit without calibrating at the cal weight prompt. This can be a handy troubleshooting feature.)

IZ set Set initial zero. If you have to ask, you probably don't need to be doing it.

lb units Enable or disable the pound units display. If enabled, the units key will allow the item weights to be displayed in pounds.

kg units Enable or disable the kilogram units display. If enabled, the units key will allow the item weights to be displayed in kilograms.

g units Enable or disable the gram units display. If enabled, the units key will allow the item weights to be displayed in grams.

oz units Enable or disable the ounce units display. If enabled, the units key will allow the item weights to be displayed in ounces.

Address Communications address. Appears as part of the total and parameter information transmitted to the Palm through the IR port.

Cntst (enabled/disabled) Controls the three key "quick contrast" adjustment feature. Default setting is "enabled" to allow adjustment after a firmware upload. It's best to disable this feature once the display contrast has been set to prevent accidental adjustment. Disabling this feature does not affect the bargraph contrast adjustment inside the setup menu.

Defaults Set all settings back to factory defaults. Will prompt for confirmation before doing its business. "Sure?" touch the Units/Cancel key to abort, or the Print/Enter key to restore factory default settings. Defaulting will erase any changes to the parameters that have been made, so print them out before defaulting just in case!

11.4 Info menu

ADC Live display of current raw unfiltered counts from analog to digital converter. Full scale is +/- 8 million counts, so values of more than several million are suspicious and could indicate a defective load cell or incorrect wiring.

Offset Current zero offset, in counts. Will update during an autozero cycle.

App Firmware application name (*tote_check_4*). Very handy to WeighTech personnel if you call and expect to receive accurate advice.

Build Firmware revision code (*Build 37*). Again, this is something that's good to know when contacting WeighTech with questions.

Date Date that firmware was compiled (*01/23/2017*). This is not the current date!

Time Timestamp of firmware compilation (*09:59:51*). This is not the current time!

Batt Current battery or power supply input voltage. Will generally be between 6 and 8 volts.

S1+ Current load cell positive signal voltage. Should be about half of the excitation voltage, and very close to the negative load cell signal voltage. (Normal range is 2 to 2.5VDC.)

S1— Current load cell negative signal voltage. Should be about half of the excitation voltage, and very close to the positive load cell signal voltage. (Normal range is 2 to 2.5VDC.)

Excite Current excitation supply voltage, updated at about 1 Hz. Should be in the 4-5 volt range for normal operating. Very low readings could indicate cable damage, incorrect load cell wiring, or an excessive number of load cells connected to the indicator.

Deadload Display current calculated deadload weight, in pounds. Assumes that the load cell outputs exactly zero counts at no load, so consider this an approximation to the actual deadload.

232 audit Experimental RS-232 output of audit trail. WeighTech use only.

IZ autoset If enabled, the scale will attempt to set an initial zero every time it's powered on. Not really relevant to this firmware revision.

IZ Current initial zero setting.

Bootload Enter bootloader for firmware update procedure. WeighTech use only.

12 Troubleshooting

12.1 Load cells

Go to the "Info menu" and verify that the "Excite" voltage is about 4.5V. A reading of less than 1V probably indicates a short from excite to ground. Confirm by removing the load cell connections. If the excite voltage reads normal with the load cell disconnected, you've got a short in the cable or a bad load cell.

Check to see that the signal voltage in the "Info menu" are about half of excite and equal. If one signal voltage is near zero, or near 4V, you may have a disconnected signal wire. Check that connection at the interface board. If the signal voltages are not near zero or 4V, but are more than a 0.5V different, you may have the load cell miswired, or a bent load cell.

If the indicator constantly shows "OVERLOAD" or "UNDERLOAD", follow the instructions above. In addition, go to the "Info menu" and watch the "ADC" reading (raw counts). It shouldn't vary more than 100-300 counts with a good load cell and a stable environment. With no load on the cell, it should be within +/- 10,000 counts of zero. (Deadload can cause the no load reading to shift.) If the no load reading is really large (say, greater than one million counts or less than negative one million counts) and the connections are solid, you probably have a bent load cell.

Unstable or noisy weights? Perform all the steps listed above. A really good test is to temporarily disconnect the load cell and substitute a known good load cell simulator (available for purchase from WeighTech), or a known good load cell. Calibrate the scale with a convenient test weight and check to see if the weight reading is stable. If so, the noisy load cell has probably been damaged or water-soaked. If the indicator still displays a noisy weight with a load cell simulator, the problem may be in the indicator. Contact WeighTech for further assistance.

12.2 Banner opto sensors

The Banner opto sensors will act strangely if wired incorrectly. Be certain that you connect the brown wire (12V) last when rewiring sensors under power. If the brown and blue wires are swapped, or if the blue wire (ground) is connected last, the sensor will go into alarm mode. In alarm mode, the sensor will appear to work but the MicroWeigh controller will not receive the correct signal. If the led indicators on the Banner appear to be working correctly, but the MicroWeigh front panel indicators don't show any changes when the beam is blocked, check the sensor wiring.

12.3 Before calling WeighTech...

Write down a few key pieces of information. Gather the indicator serial number from the front panel, the firmware application name and build number from the "Info menu", and grab the current settings if you have access to a Palm. If anything on the indicator has changed, been replaced, or been modified, mention that to the service technician too. If the problem involves fill rates, hangs, or questions regarding machine capabilities, be ready to describe the product, product flow rate, and any bag/box/tote/combo sizes. If you're calling about unstable weight readings, over/underload, or other load cell related problems, have the ADC, excite, and signal readings from the "Info menu" handy. When calling, be prepared to describe what is wrong ("it doesn't work!" isn't a good description–"hopper gate doesn't shut in off mode" is much better) and what you expected to see.

12.4 Module assignments

Slot	Module Type	Function	Module on (lit)	Module off (dark)
M1	OAC5 (black)	Box stop	Up	Down
M2	OAC5 (black)	Weighbridge	Up	Down
M3	OAC5 (black)	Knockoff	Up	Down
M4				
M5				
M6	IDC5 (white)	Opto box sense	No box	Box

13 Replacement Parts

Part Number	Description
1000-10	Load cell, 200 lb capacity
AA0009	Air regulator
AA0014	Rear air cylinder mount
AA0015	Air cylinder rod clevis
AC0003	Air cylinder, stop gate
AC0007	Tote lift air cylinder
AV0044	Air valve and coil assembly (Festo)
AV0045	DIN plug, air valve assembly (Festo)
AV0046	LED gasket, air valve assembly (Festo)
AV0047	Substation, air valve assembly (Festo)
AV0048	End plate kit, air valve assembly (Festo)
CB0049	Belting, series 3000, knuckle chain
EF0009	Strain relief, MicroWeigh
EP0020	Van Der Graaf motor, 4.5" dia., 14.17" length, 80 FPM
HW0018	MicroWeigh housing screw (pack of 4)
HW0019	Screw, 6-19 x 0.375, trilobe PPH, MicroWeigh (pack of 10)
SE0035	Photo electric receiver, detachable (T18SP6RQ)
SE0036	Photo electric emitter, detachable (T186EQ)
SE0037	Cord set, straight connector (MQDC-430)
SR0001	Input relay (white, G4-IDC5)
SR0002	Output relay (black, G4-OAC5)
WE0028	Main gasket, MicroWeigh
WE0029-1	Power cord, detachable, MicroWeigh
WE0046-101	Front housing assembly, MicroWeigh machine control (specify firmware name and revision)
WE0047-101	MicroWeigh machine control back housing (no board)
WE0047-102	MicroWeigh machine control back housing (with board)
WE0076-101	Machine control interface board assembly, MicroWeigh
WF1300-101	Tote weigh lift kit

JOB: TOTE CHECKER

PLANT:

LOCATION:

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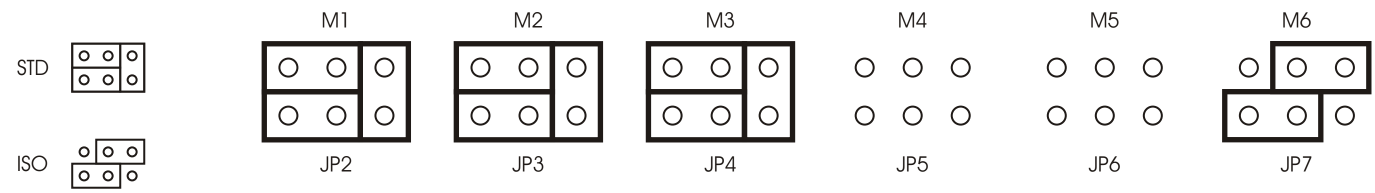
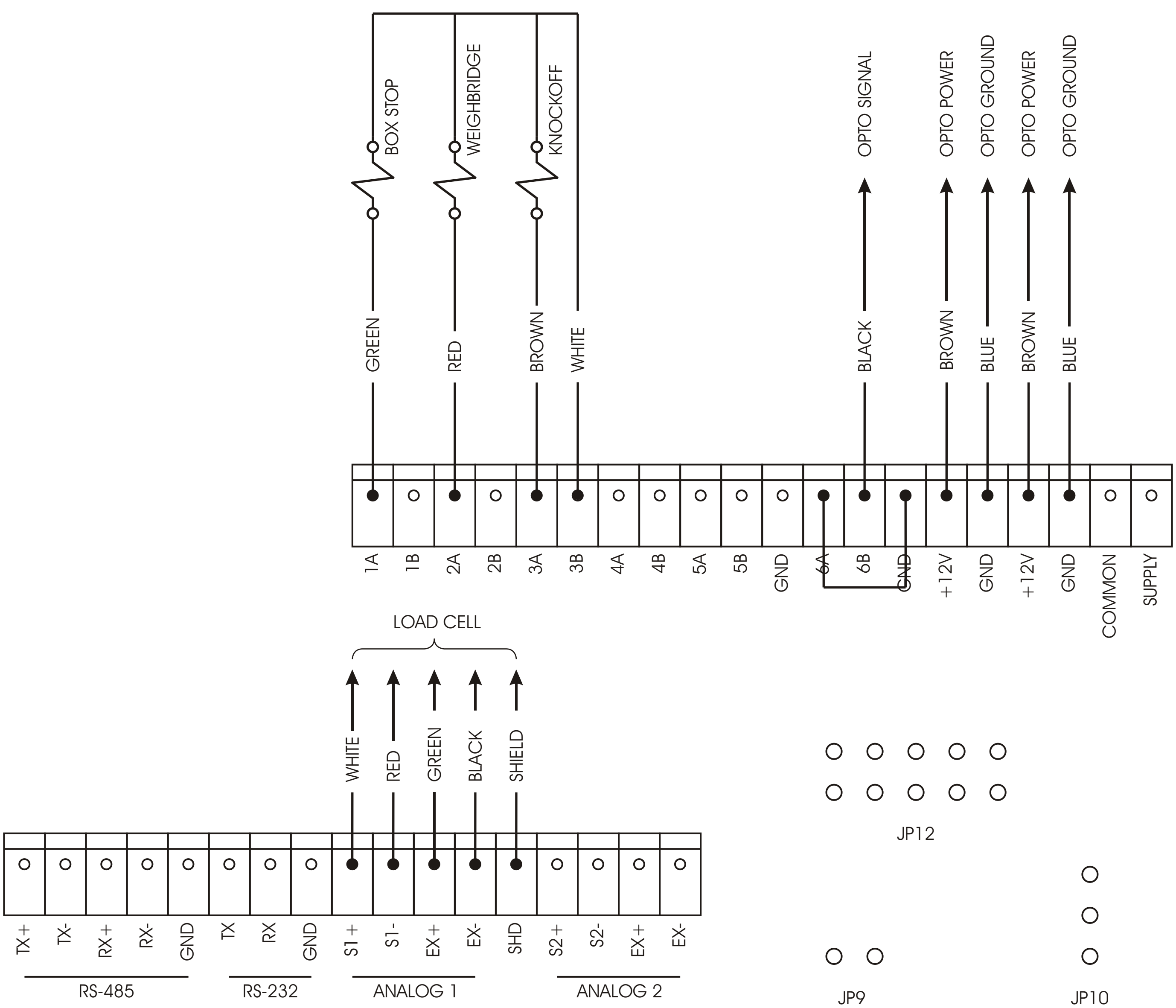
DRAWN BY: NEWELL

DATE: 8-29-2018

FIRMWARE: TOTE_CHECK_4



SLOT	TYPE	FUNCTION	LIT	DARK
M1	BLACK OAC-5	BOX STOP	UP	DOWN
M2	BLACK OAC-5	WEIGHBRIDGE	UP	DOWN
M3	BLACK OAC-5	KNOCKOFF	UP	DOWN
M4				
M5				
M6	WHITE IDC-5	BOX SENSOR	NO BOX	BOX



15 Load Cell Color Codes

Manufacturer	*Models*	*Signal +*	*Signal -*	*Excite +*	*Excite -*	*Shield*	*Sense +*	*Sense -*
Advanced Transducers		Green	White	Red	Black	Bare wire		
Allegany Technology		Red	White	Green	Black	Bare wire		
Artech		Green	White	Red	Black			
Beowulf		White	Red	Green	Black			
BLH	C2P1 C3P1 T2P1 T3P1	White	Red	Green	Black	Yellow		
Cardinal		White	Red	Green	Black	Bare wire		
Celtron	CSB DSR LOC SQB STC STC-SS DSR CLB HED DLB	Green	White	Red	Black	Bare wire		
	LPS	Green	White	Red	Blue	Bare wire		
	HOC MOC	Red	White	Green	Black	Bare wire		
Dillon	Canister Tension	Black	Red	Green	White	Orange		
	Compression	Black	Red	White	Green	Orange		
	Z-cell	White	Green	Red	Black	Orange		
Force Measurement		Green	White	Red	Black	Bare wire		
GSE		White	Green	Red	Black	Bare wire		
HBM	BLC BLF JRT PWS RSC SBF SB3 USB U1T Z6	White	Red	Green	Black	Yellow		
	BBS	White	Red	Green	Black	Bare		
	PLC B35	Green	White	Red	Black	Yellow		
	SP4	White	Red	Green	Black	Yellow	Orange	Blue
Interface	SSM 1200 3200	Green	White	Red	Black	Bare wire		
Kubota		Green	Blue	Red	White	Yellow		
National		White	Red	Green	Black	Yellow		
NCI		White	Green	Red	Black	Bare wire		
Pennsylvania		Green	White	Orange	Blue	Bare wire		
Phillips		Green	Grey	Red	Blue	Bare wire		
Revere Transducer	62HU 63HU 363 953 9523	Green	White	Red	Black	Bare wire		
	92CC 93CC 42U 43U 263D 462 5102 5103 5123 5223 5723 6762 9102 9103 9123 9363	Green	White	Red	Black	Orange		
	392B 642 652 692B2 BSP HPS USP1	White	Red	Green	Black	Bare wire		
	792 933 SHB SSB	White	Red	Green	Black	Clear		
	CP1 CSP1	White	Red	Green	Black	Orange		
	RLC	Brown	White	Pink	Grey	Bare wire		
Rice Lake	RL20000 RL20000SS RL20001 RL20001HE RL30000 RL35023 RL35023S RL35082 RL35082S RL35083 RL39123 RL39523 RL50210 RL65044 RL70000 RL75016 RL75016SS RL75040A RL75058 RL75060 RL75223 RL90000 RLETB RLETS RLHSS RLMK4 RL50500 RL70000SS RL71000HE RL75016HE RLMK15 RLMK21 RL75061	Green	White	Red	Black	Bare wire		
	RLMK1	White	Green	Red	Black	Orange	Yellow	Blue
	RL1521	Green	White	Red	Blue	Bare wire		
Sensotec	White	Green	Red	Black	Bare wire			
Sensortronics	60001 60008 60018 60030 60036 60040 60048 60048SS 60050 60051 60060 60060-0101 60063 65007 65016 65016SS 65016W 65023 65023S 65023SS 65024 65040A 65040S 65058 65058S 65061A 65083 65083S 65114	Green	White	Red	Black	Bare wire		
	60007 60064	White	Red	Green	Black	Bare wire		
	65088-1000 65088-1114	White	Red	Green	Black	Orange		
Tedea Huntleigh	4158	Green	White	Red	Black	Bare wire		
	3411 3421	Green	White	Red	Black	Bare wire	Blue	Brown
	240 1010 1022 1040 1042 1140 1250 1260 1320 9010 605 1030 1240 1241	Red	White	Green	Black	Bare wire	Blue	Brown
	355 620 3510	White	Red	Blue	Black	Bare wire	Green	Grey
Toledo		White	Red	Green	Black	Yellow		
Weigh-Tronics		White	Red	Green	Black	White/Orange		

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